

Bengen's work pointed out that sequence-of-returns risk will reduce safe, sustainable withdrawal rates below what is implied by the average portfolio return. Its popularity has coalesced into what we are calling the probability-based approach. It will be the focus of this book.

Next, we'll discuss probability-based answers to some basic retirement income planning questions.

How are goals prioritized?

The idea of using a safe withdrawal rate, as implied by the 4 percent rule, is that people don't retire until they have accumulated a sufficient level of assets that can meet their entire lifestyle goal by spending from their portfolio at the determined safe withdrawal rate. For instance, if someone seeks to spend \$40,000 per year from his or her portfolio and is comfortable with spending at an initial 4 percent rate from assets, then the wealth accumulation target to allow retirement to commence is:

$$\text{Wealth} = \text{Spending} / \text{Withdrawal Rate} = 40,000 / 0.04 = \$1,000,000$$

According to probability-based advocates, people identify lifestyle spending needs that must be met to fulfill the standard of living they have in mind for retirement. If they are unable to meet these lifestyle spending goals, they will view their retirement as a failure. Thus, the emphasis is on minimizing the probability of failure (or, conversely, maximizing the probability of success) for the overall lifestyle goal without concern for the potential magnitude of those failures when they happen. If retirees are generally more sensitive to the probability of meeting their goals than to the magnitude of their shortfall, then it hardly matters if they can spend only one-quarter or one-half less than their goal, because their lifestyle is severely diminished either way.

As suggested by the naming of the probability-based school, the objective is to develop a plan that will maximize the probability of success for meeting the overall lifestyle goal. For aggressive goals, an aggressive asset allocation may maximize success with the hope that an outsized return premium for stocks can be earned above bonds. Financial planners such as Michael Kitces and Jonathan Guyton argue that it is difficult for people to